Cycle de Conférence: Simulating and analyzing order book data using the queue-reactive model

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1 Introduction

The Limit Order Book (LOB) is the central mechanism of modern electronic financial markets. It represents the aggregate supply and demand for a specific asset at a given time. The state of the LOB is determined by the interaction of three primary types of order flows:

- Limit Orders (LO): Orders to buy or sell a specified quantity at a specific price. These orders provide liquidity and sit in the queue until executed or cancelled.
- Market Orders (MO): Orders to buy or sell immediately at the best available price. These orders consume liquidity and trigger immediate trades.
- Cancellations (C): The withdrawal of previously posted limit orders.

In this framework, the dynamics of the best bid and best ask prices are driven by the depletion of the queues at the best limits. When the queue at the best ask (resp. best bid) is fully depleted by market orders or cancellations, the price moves up (resp. down).

We give here an compact resume of the paper from Weibing Huang, Charles-Albert Lehalle, and Mathieu Rosenbaum on the models for simulating order books [1].

2 General Framework and Definitions

We consider a single asset traded in a purely electronic market organized through a LOB. Time is continuous and denoted by $t \geq 0$.

2.1 Price Grid and Tick Size

Prices lie on a discrete grid: $\mathcal{P} = \{k\delta \ : \ k \in \mathbb{Z}\}$ where $\delta > 0$ is the tick size. At time t, we denote: P_t^b (resp P_t^a) the best bid price (resp best ask price), with spread $S_t = P_t^a - P_t^b \geq \delta$. The relevant reference price p_{ref}, which provides the center point of the LOB and, is defined by :

$$p_{ref} = \frac{P_t^a + P_t^b}{2}$$

2.2 Queue Representation

At each price level $p \in \mathcal{P}$, the book contains a queue of limit orders. We consider K price levels on each side of the book, indexed by

$$\{-K, \ldots, -1, 1, \ldots, K\},$$

where Q_{-i} denotes the i-th bid queue (left of p_{ref}) and Q_i the i-th ask queue (right of p_{ref}). The queue size, in number of unit orders, at level i is denoted by $q_i(t) \in \mathbb{N}$. The LOB state during such a period is thus the $2K$-dimensional vector

$$X(t) \ = \ (q_{-K}(t), \ldots, q_{-1}(t), q_1(t), \ldots, q_K(t)) \in \Omega := \mathbb{N}^{2K}.$$

In this regime, $X(t)$ is modeled as a continuous-time Markov jump process.

Between price changes, the reference price is fixed and only queue sizes evolve. When a best queue hits zero, a price jump occurs and the book is shifted accordingly. This Markovian structure is the foundation for both the following models.

3 Markov Jump Process Model

3.1 Assumptions (order flows and Markov property)

The model relies on the following structural assumptions: (1) over the considered time window, p_{ref} does not move. Only the queue sizes evolve. (2) Each elementary event (limit order insertion, cancellation, market order consumption) modifies exactly one queue q_i by ± 1. (3) Conditional on the current book state $q \in \Omega$, the next event occurs with intensities that depend only on q (Markov property). In particular, arrivals at level i have intensity $f_i(q)$ and departures at level i have intensity $g_i(q)$.

3.2 Infinitesimal generator

Let e_i denote the canonical basis vector in $\mathbb{R}^{2K}$ corresponding to coordinate i (i.e. $(e_i)_j = \mathbf{1}_{\{j=i\}}$). For any state $q = (q_{-K}, \ldots, q_{-1}, q_1, \ldots, q_K) \in \Omega$, the infinitesimal generator Q of the

Markov jump process $X(t)$ is specified by:

$$Q_{q,q+e_i} = f_i(q), \tag{1}$$
$$Q_{q,q-e_i} = g_i(q), \tag{2}$$
$$Q_{q,q} = -\sum_{p \neq q} Q_{q,p}, \tag{3}$$
$$Q_{q,p} = 0 \quad \text{otherwise.} \tag{4}$$

The interpretation is direct: $q \to q \pm e_i$ corresponds to a limit order insertion or liquidity removal at level i of one unit.

3.3 Stability and ergodicity conditions

To ensure the model is well-posed and admits a stationary regime during constant-p_{ref} periods, Huang–Lehalle–Rosenbaum introduced two general assumptions.

- *Assumption 1* There exist $C_{\text{bound}} \in \mathbb{N}$ and $\delta > 0$ such that for all i and all $q \in \Omega$, if $q_i > C_{\text{bound}}$, then $f_i(q) - g_i(q) < -\delta$. Intuitively, when a queue becomes very large, departures dominate arrivals, preventing runaway growth.
- *Assumption 2* There exists $H > 0$ such that for any $q \in \Omega$, $\sum_{i \in \{-K,\dots,-1,1,\dots,K\}} f_i(q) \leq H$. So that total arrival intensity remains uniformly bounded, preventing explosion of event times.

Theorem 1 (Ergodicity of the LOB dynamics)**.** *Under Assumptions 1 and 2, the $2K$-dimensional Markov jump process $X(t)$ is ergodic: it admits an invariant distribution π, and for any $q, p \in \Omega$,*

$$\lim_{t \to \infty} \mathbb{P}\big(X(t) = p \mid X(0) = q\big) = \pi(p).$$

3.4 Example: Model 1

In Model I of [1], the LOB around a fixed reference price is modeled as a collection of independent queues. At each level i, three order flows are considered: limit orders, cancellations, and market orders. Their conditional intensities depend only on the local queue size q_i, and the flows are assumed independent across levels. Denoting the intensity functions by $\lambda_i^L(\cdot)$ (limit), $\lambda_i^C(\cdot)$ (cancel), and $\lambda_i^M(\cdot)$ (market), we set

$$f_i(q) = \lambda_i^L(q_i), \qquad g_i(q) = \lambda_i^C(q_i) + \lambda_i^M(q_i), \tag{5}$$

so that each coordinate $q_i(t)$ is a birth–death process with birth rate $\lambda_i^L(q_i)$ and death rate $\lambda_i^C(q_i) + \lambda_i^M(q_i)$. Moreover, by the symmetry assumption of the LOB we have identical intensity functions for i and $-i$, $\lambda_{-i} = \lambda_i$.

Under the usual positive-recurrence condition, the invariant distribution of each queue can be written in closed form. Define the arrival/departure ratio (for $n \geq 0$) by

$$\rho_i(n) = \frac{\lambda_i^L(n)}{\lambda_i^C(n+1) + \lambda_i^M(n+1)}. \tag{6}$$

Then the stationary law π_i of q_i satisfies;

$$\pi_i(n) = \pi_i(0) \prod_{j=1}^{n} \rho_i(j-1), \qquad \pi_i(0) = \left(1 + \sum_{n=1}^{\infty} \prod_{j=1}^{n} \rho_i(j-1)\right)^{-1}. \tag{7}$$

Since queues are independent in Model I, the invariant distribution of the LOB factorizes as

$$\pi(q) = \prod_{i \in \{-K,\dots,-1,1,\dots,K\}} \pi_i(q_i).$$

Hence, the long-run shape of the book is fully determined by the ratio functions ρ_i: different assets may have different raw flow intensities, yet exhibit the same stationary LOB profile if they share the same arrival/departure ratios.

4 The Queue-Reactive Model

The objective of the Queue-Reactive model is to obtain a time-consistent LOB model on long horizons by coupling: (i) a Markovian queueing dynamics when the reference price is fixed, and (ii) a stochastic dynamics for the reference price.

4.1 Assumptions

- Between two changes of p_{ref} at time t, the vector of queue sizes $X(t)$ evolves as in Section 3 with same generator.
- The model introduces reference price updates. When an event modifies the midprice p_{mid}, the reference price p_{ref} moves by one tick δ with probability θ, provided the corresponding best queue is empty at that moment ($q_{\pm 1} = 0$).
- When p_{ref} changes, the whole LOB state is redrawn from its invariant distribution around the new reference price.

Under these assumptions, the market is modeled by the $(2K+1)$-dimensional Markov process

$$\tilde{X}(t) := \big(X(t),\, p_{\text{ref}}(t)\big),$$

taking values in a countable state space $\mathbb{N}^{2K} \times \mathcal{P}$.

4.2 Dynamics when p_{ref} changes

A change of p_{ref} can be triggered by any of the following events:

- insertion of a limit order inside the spread when the opposite best queue is empty (e.g. buy inside-spread while $q_1 = 0$);
- cancellation of the last order at one best queue;
- market order consuming the last order at one best queue.

When p_{ref} increases (resp. decreases), each queue q_i switches immediately to the value of its neighbor to the right (resp. left). In particular, after a move, one of the new best queues becomes empty ($q_{\pm 1} = 0$ depending on the direction).

We then have multiple effects:

Boundary refresh: Since the book is only tracked up to level K, when p_{ref} moves, the newly created outer queue (e.g. $q_{\pm K}$) is drawn from its invariant distribution.

Renormalization by average event size: Queue switching must account for the fact that average event sizes differ across levels: when q_i takes the value of q_j, it should be rescaled by the ratio of the corresponding average event sizes.

Reinitialization: Finally, at each reference price change, with probability θ_{reinit} the whole vector $X(t)$ is redrawn from the invariant distribution around the new p_{ref} where the paper interprets θ_{reinit} as the fraction of price moves driven by exogenous information.

5 Simulation of the Queue-Reactive Limit Order Book

We now describe the simulation procedure used for the queue-reactive limit order book introduced above. The objective is to generate realistic bid/ask dynamics from state-dependent order flow intensities. Code can be found at : `https://github.com/Obrks/queue_reactive_model_HFT/tree/main`

5.1 State-Dependent Intensities

The queue dynamics are driven by state-dependent intensities for limit insertions, cancellations, and market orders. For each queue i, we define:

$$\lambda_i(q) \quad \text{limit order arrivals}, \qquad \lambda_i^C(q) \quad \text{cancellations}, \qquad \lambda_i^M(q) \quad \text{market orders}.$$

The total removal intensity is then given by, $\mu_i(q) = \lambda_i^C(q) + \lambda_i^M(q)$. The implemented specification distinguishes between three cases: the best queues $Q_{\pm 1}$, the second queues $Q_{\pm 2}$, and deeper queues $Q_{\pm j}$ for $j \geq 3$.

Market orders intensities depends on the size of the corresponding best queue. These market orders may consume the entire best queue and, if their size exceeds the available liquidity at the touch, they walk further into the book.

The best queues have highly state-dependent intensities, reflecting the empirical fact that liquidity at the touch behaves differently from liquidity deeper in the book. The second queues are also treated separately, since they behave differently whenever the best queue is depleted or not. Finally, deeper queues are assigned more stable insertion and cancellation intensities.

5.2 Queue Update Rule

At each simulation step, for each queue i, arrivals and removals are sampled according to Poisson random variables:

$$N_i^A \sim \text{Poisson}(\lambda_i(q)\,\Delta t), \qquad N_i^R \sim \text{Poisson}(\mu_i(q)\,\Delta t),$$

where Δt denotes the time step. The queue size is then updated by

$$q_i \leftarrow \max\left(0,\, q_i + N_i^A - N_i^R\right).$$

We also introduce additional controls to prevent queues from growing unrealistically large. If a queue exceeds a threshold C_{bound}, a penalty term is applied to reduce excessive accumulation. Moreover, the total incoming flow is capped by a parameter H, to limit simultaneous growth across many queues. They can be parameterized in the code for different market behaviors.

5.3 Dynamic reference price

A key mechanism of the model is that the reference price moves whenever the best queue is depleted. If the best bid/ask queue Q_{-1}/Q_1 is emptied, the bid/ask side is shifted, a new far bid/ask queue is inserted, and the reference price decreases by half a tick:

$$p_{\text{ref}} \leftarrow p_{\text{ref}} \pm \frac{\delta p}{2}.$$

We preserve exactly k queues on each side of the book while generating price changes through queue depletion.

5.4 Simulation Algorithm and Features

The simulation is implemented through an `OrderBook` class initialized with a reference price p_{ref}, a list of queue states, and a book depth k. From these inputs, the class constructs the price grid, stores the state of each queue, and initializes the state-dependent intensity functions governing insertions, cancellations, and market orders.

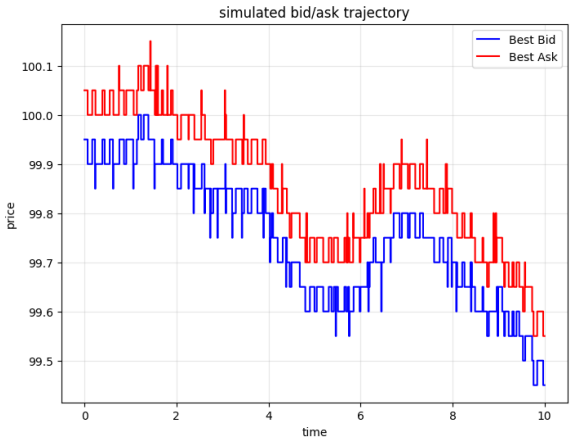
At each simulation step, the method `update_state` updates all queue sizes by sampling Poisson arrivals and removals according to the current state of the book. The method `handle_depletions` then checks whether the best bid or best ask queue has been emptied; if this happens, the corresponding side of the book is shifted inward, a new far queue is inserted, and the reference price is adjusted. Next, the method `process_market_orders` simulates explicit market orders at the best quotes and executes them through the book.

Repeating this step over a time grid generates a trajectory of the simulated bid–ask process. From this simulated path, one can then produce graphical representations of the order book shape, the bid and ask trajectories, the spread dynamics, or other summary statistics.

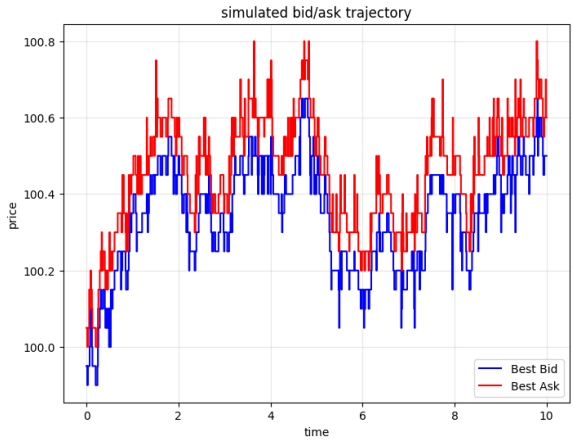
The resulting simulated order book reproduces several stylized features: the best bid and best ask evolve as piecewise constant processes, the mid-price changes by half-tick when one side of the book is depleted, the spread may widen temporarily and can be controlled, deeper queues remain more stable than the first few queues. This produces realistic short-term bid–ask dynamics which can be easily parameterized and modified.

5.5 Graphs

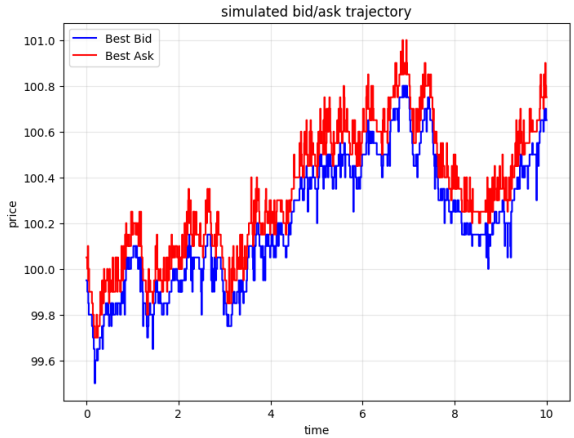
To illustrate the effect of the simulation parameters on the bid–ask dynamics, we consider three different regimes. First, a low-activity regime which produces relatively stable trajectories, with fewer queue depletions, limited price variation and with bigger spreads. Second, an intermediate regime which yields a more balanced behavior, where price changes occur more frequently while the book remains reasonably stable. Finally, a high-activity regime which generates a much more reactive order book: best queues are depleted more often, the reference price moves more frequently, and smaller spread generally.



(a) Low-activity regime
$(\texttt{stf}, \delta, H) = (5, 1.0, 20)$



(b) Intermediate regime
$(\texttt{stf}, \delta, H) = (10, 0.5, 10)$



(a) High-activity regime
$(\texttt{stf}, \delta, H) = (20, 0.2, 5)$

6 Conclusion and limitations

The models in [1] offer a simple, interpretable and efficient framework to simulate a limit order book: a finite set of queues evolving as a Markov jump process, with a queue-reactive extension to generate price moves over long horizons.

One of the main strengths of this framework is its balance between realism and simplicity. The model captures the dependence of order flow on the local state of the book, distinguishes the different queue levels, and provides an intuitive mechanism for bid–ask dynamics, spread changes, and reference price updates. Well suited for simulation studies and as a first approximation of high-frequency order book behavior.

However, the approach remains highly stylized. First, the Markov assumption makes the dynamics memoryless, whereas real order flow exhibits strong persistence, clustering, and long-range dependence. Second, the book is represented with a finite depth, so the treatment of boundary queues and reinitialization is necessarily artificial. Third, the model captures only limited interaction effects between price levels, order types, and market participants, and therefore cannot fully account for strategic behavior such as order splitting, hidden liquidity, or anticipatory cancellations.

A natural alternative is to model order arrivals with Hawkes processes, where intensities depend on the past event history, capturing self-excitation between order types and book sides as well as the clustering that is characteristic of high-frequency financial data. They provide a better statistical fit and more realistic simulations for the order book. At the state of the art level, recent approaches based on neural point processes or deep-learning-based intensity models offer even greater flexibility and can capture highly nonlinear interactions in the order flow.

The trade-off is that these richer models are significantly harder to calibrate, interpret, and simulate. As a result, the queue-reactive framework remains a valuable benchmark.

References

[1] Weibing Huang, Charles-Albert Lehalle, and Mathieu Rosenbaum. Simulating and analyzing order book data: The queue-reactive model. *Journal of the American Statistical Association*, 2014.